**18** Fig. 1 shows how the displacement of a wave varies with distance along the wave at one particular time.

Fig. 2 shows how the displacement of one particular point P of the same wave varies with time.

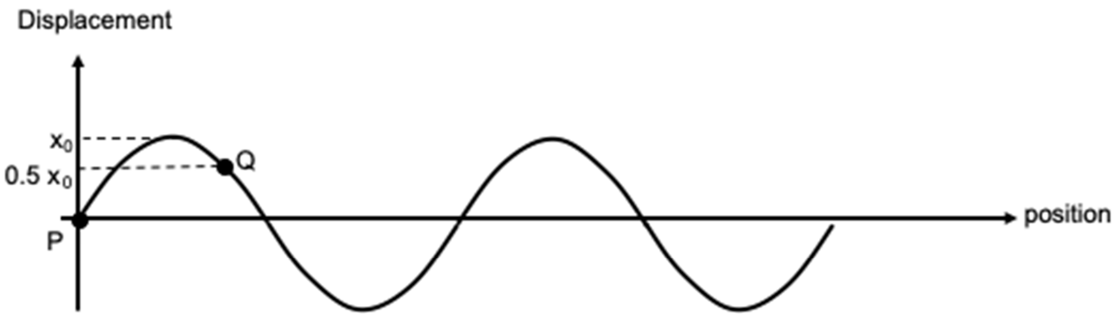



**Fig. 1**

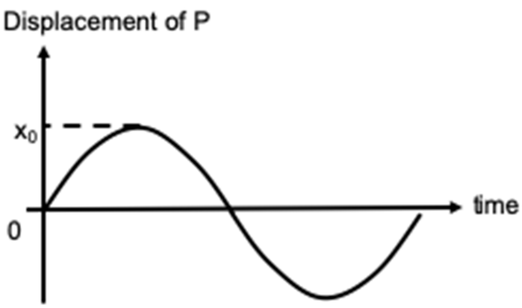
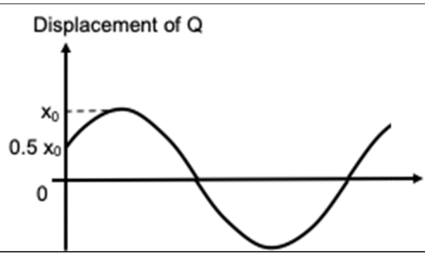


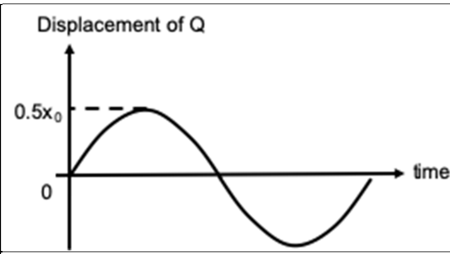

**Fig. 2**

Which of the following graphs correctly depicts the displacement-time graph for point Q?

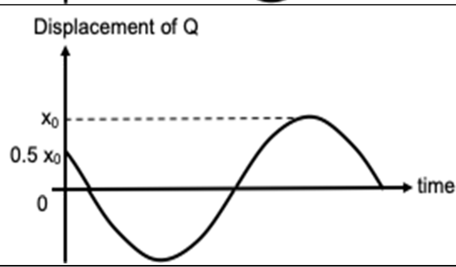
**A**



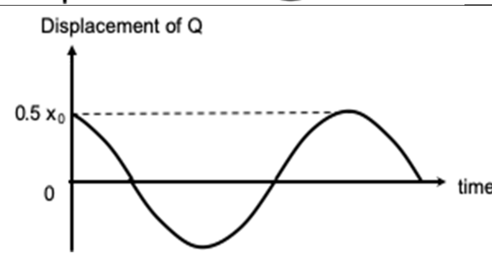

**B**




**C**

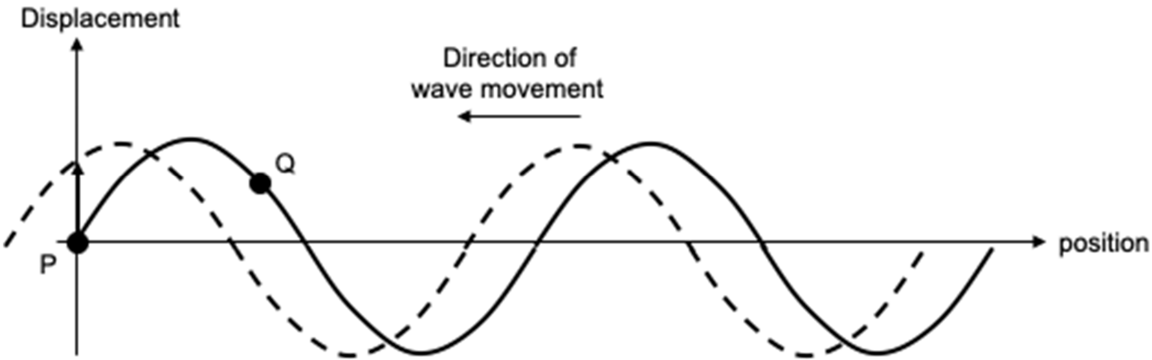



**D**




**Answer: C**

From Fig. 2, we can deduce that the wave is moving in the leftwards direction, as depicted below.




If the wave is travelling to the left, its next position can be represented by the dotted line shown above. A can be seen to be moving from equilibrium to a displacement in the positive direction, and B is moving downwards.

As such, Option A is not correct. In the same light, Option B is also not correct.

Since P and Q are succumbed to the same wave, P and Q will encounter the same amplitude magnitude, which is $x_0$. Therefore, Option D is wrong.